Packaging the Splittings

Realizability, collection monads, and the choice content of the McBride derivative in a finite-limits metalanguage

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Abstract

i want to make visible the seams along which a logic is assembled. The commitment that makes the seams visible is a realizability semantics: a formula denotes the collection of computations that witness its computational content. Under that commitment a logic factors into three components — a *collection* component (how witnesses are aggregated), a *structural* component (term constructors lifted to predicates on witnesses), and a *behavioral* component (contexts labeling transitions lifted to modalities). The collection component sits awkwardly with categorical logic, because the subobject classifier bakes in *set*-shaped collection: there is no list of all lists. i reconcile the two by going syntactic. Treating every collection as a term type, i use the McBride derivative to form the object of all splittings $(\partial T, T)$ — the *proto power object* — and then ask what must be added to the ambient metalanguage to *package* the splittings back into the collection type. Committing the metalanguage to finite limits (cartesian logic) is what keeps the whole apparatus syntactic: there is no subobject classifier to lean on, so the derivative route is the only route; and choice cannot be asserted as a proposition, so it can enter only as a named section operation. The central claim is that packaging is exactly a section of the quotient presenting the collection monad, that its choice content equals the assertion that the atom object admits a linear order, and that going syntactic discharges this because term positions are already ordered. The construction is kept parametric in the order on atoms, isolating the choice residue at the atom boundary — which is where later reflective (knot-tying) constructions will act. Throughout i note that the equations one quotients by — commutativity and idempotence — are exactly the equations the no-go theorems for distributive laws turn on, so the distributive-law obstruction and the packaging cost are one datum read twice.

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1 Why realizability, and why this matters for the argument to the AI community

Most accounts take a logic as given: a fixed vocabulary of connectives, a notion of model, and a truth predicate that floats above computation. Truth values arrive from a Boolean or Heyting algebra that is postulated, not built. For a discipline that wants its reasoning systems to be inspectable — traceable to constructions whose behavior can be run, audited, and held accountable — this is exactly the wrong starting point. The glue that binds truth to the algebra is opaque, and once it is in place one can no longer see what the logic is *made of*.

i want the opposite. i want a logic whose every connective is an operation on computations, so that the entire edifice is a construction and nothing is postulated except the constructions themselves. The commitment that delivers this is realizability. Rather than asking whether a formula is true, we ask which computations *witness* it, and we let the formula denote the collection of its witnesses:

$$[\![\varphi]\!] \;=\; \{\, e : e \Vdash \varphi \,\}.$$

This is the Kleene move [1, 2], but with the aggregating container promoted from an incidental detail to a *parameter* of the construction. Once witnesses are computations and formulae are collections of them, every connective is forced to be an operation of one of exactly three kinds, corresponding to the three things one can vary: how witnesses are collected, how they are built, and how they move. This is the decomposition i develop in §2.

There is a second, sharper commitment, and it is the one i most want to defend to the AI community. i insist that the ambient metalanguage — the language in which the collections, the constructions, and the packaging all live — be a *finite-limits* (cartesian, essentially-algebraic) theory [12, 13, 14]. The reason is not minimalism for its own sake. It is that a finite-limits metalanguage is the maximal commitment to "everything is a construction":

- there are no truth values floating in from a Boolean algebra, because there is in general no subobject classifier (a lex category need not be a topos);

- there is no non-constructive existential, because the only existentials one may assert are provably *unique* ones, carved out as partial operations on equationally-defined subdomains;
- and — the point that will do the real work in §6 and §8 — choice cannot be *stated* as a proposition at all. There is no $\forall$, and $\exists$ is available only when the witness is unique. So the only way to "add choice" is to add a *function symbol* to the signature, together with equations making it a section. In a finite-limits metalanguage, choice is not a truth one assumes; it is an operation one names.

This is exactly the inspectability i am after. When choice enters, it enters as a symbol you can point at, with a documented arity and documented equations, and you can localize precisely where in the construction it was spent. Nothing is hidden in a semantic ether. The seams are visible because we have refused the glue — the subobject classifier — that ordinarily hides them.

The remainder of the note carries this out. §2 gives the three-component decomposition. §3 shows that the collection component is a substructural dial whose settings are governed by two equations, commutativity and idempotence. §4 states the disconnect with categorical logic, and §5 shows that the same two equations are what the no-go theorems for distributive laws turn on — so the obstruction to a uniform collection component and the cost of packaging are, as we will see, one datum read twice. §6 cashes out the finite-limits commitment. §7 builds the proto power object from the McBride derivative, and §8 defines packaging as a section and states the choice conjecture, parametric in the order on atoms. §9 places the conjecture in the Weihrauch lattice, §10 works the three instances, and §11 draws out the consequences, including the absence of any classicality collapse and the reflective use of the atom-order parameter.

2 Three components of a logic under realizability

Fix a representation of computation. Throughout i take it to be a term rewrite system: terms are computational states, and rewrites are steps. Write T for the type of terms. Fix also a way of aggregating witnesses: a *collection type* S, which will always carry the structure of a monad (§3). A formula denotes an object $[\![\varphi]\!] : S(T)$ — a collection of terms.

Under these two fixings, three families of connectives arise, and every connective is of one of the three kinds.

Collection connectives. Any operation on S lifts to a connective. If witnesses are collected in sets, the semilattice operations of $S = \mathsf{P}_{\text{fin}}$ give the familiar connectives:

$$[\![\varphi \wedge \psi]\!] = [\![\varphi]\!] \cap [\![\psi]\!], \qquad [\![\varphi \vee \psi]\!] = [\![\varphi]\!] \cup [\![\psi]\!].$$

This is where the choice of container is felt: change S from sets to multisets, lists, trees, or distributions, and these connectives change with it. The collection component is the parameter i most want to vary, and §3 explains what varying it does.

Structural connectives. Each term constructor of T lifts to a predicate transformer. If $f : T^n \to T$ is an n-ary constructor, define

$$[\![\widehat{f}(\varphi_1, \ldots, \varphi_n)]\!] \;=\; \{\, f(e_1, \ldots, e_n) : e_i \Vdash \varphi_i \,\}.$$

The lifted constructor $\widehat{f}$ is a *structural* predicate: it speaks about the *shape* of a witness, asserting that the witnessing computation was built by f from sub-witnesses of the φ_i. In the reflective setting these are the constructors of the rho calculus lifted to namespace predicates [3, 4]; in general they are the introduction half of an Operational-Semantics-in-Logical-Form (OSLF) presentation.

Behavioral connectives. Each rewrite is labeled by the context in which it fires. A step $C[l] \to C[r]$ carries the one-hole context C, an element of the *derivative* ∂T (§7). Lifting the labels to modalities gives

$$[\![\langle C\rangle\varphi]\!] = \{\, e : e \xrightarrow{C} e',\ e' \Vdash \varphi \,\}, \qquad [\![[C]\varphi]\!] = \{\, e : \forall e'.\ e \xrightarrow{C} e' \Rightarrow e' \Vdash \varphi \,\},$$

the context-indexed Hennessy–Milner modalities. This is the elimination half of OSLF.

Remark 2.1. The structural and behavioral components are the two halves of OSLF, generated respectively from constructors and from labeled transitions. They are already well-understood in the reflective setting. The component that resists the categorical-logic packaging, and that this note is about, is the first one — collection — together with the derivative machinery (∂T) that turns out to glue it to the third.

3 The collection component is a substructural dial

The collection type S is a monad, and the algebraic theory presenting S is not incidental: *its equations are the structural rules of the propositional logic the collection component induces.* This is the first fact that makes the decomposition worth having, because it says the collection component is not "logic plus a data-structure flavor"; it *is* the substructural hierarchy, indexed by how far S sits from set-shaped.

Write the binary aggregation of S as $*$ with unit ε. The correspondence is:

$$\begin{array}{lcl} \text{commutativity of } * & \longleftrightarrow & \text{exchange,} \\ \text{idempotence of } * & \longleftrightarrow & \text{contraction,} \\ \text{unit/absorption laws} & \longleftrightarrow & \text{weakening.} \end{array}$$

Reading the standard collection monads through this correspondence:

- P_{fin}, the finite-powerset monad — a commutative, idempotent monoid (a join-semilattice) — gives exchange and contraction (and, with the unit, weakening): the full structural regime, hence Heyting/Boolean propositional logic.
- Bag, the finite-multiset monad — the free commutative monoid, commutative but not idempotent — gives exchange without contraction: the multiplicative-linear regime with $\otimes$.
- List, the list monad — the free monoid, neither commutative nor idempotent — gives neither exchange nor contraction: a non-commutative, Lambek-style regime.
- D, the distribution monad — commutative with a convex (barycentric) structure in place of idempotence — gives the probabilistic regime.

Principle 3.1 (Collection is substructure)**.** *The setting of the collection dial is the pair of Booleans* (*commutative?*, *idempotent?*) *of S, and this pair is exactly the presence or absence of exchange and contraction in the induced logic.*

Keep the two coordinates of Principle 3.1 in view. They are the same two coordinates on which the no-go theorems of §5 turn, and the same two equations one quotients by when packaging in §8. Everything in this note is, at bottom, bookkeeping on commutativity and idempotence.

4 The disconnect with categorical logic

Categorical logic packages the collection component through the subobject classifier Ω: the power object of A is Ω^A, the object of subobjects of A. This is a beautiful and total solution — for one setting of the dial. It hard-wires $S =$ powerset. Two things go wrong when we try to honor Principle 3.1 and turn the dial.

First, the classifier fixes the collection shape. There is no Ω-parametrized-by-S; the power object is subsets, and only subsets. One cannot, inside the Ω story, ask for the multiset or list of witnesses.

Second, and more tellingly, even the operation of forming "the collection of all subcollections *as an object of the same shape*" fails away from sets. The slogan is: *there is no list of all lists.* To assemble all sublists of a given list into a single *list*, one must linearly order the sublists; there is no canonical order on them, and choosing one is choice. For sets the analogous assembly is free of order — the subsets form a set with nothing to arrange — which is exactly why the Ω story works for the powerset setting and only there.

Remark 4.1. The disconnect is not a defect of categorical logic to be repaired by a cleverer classifier. It is a signal that the packaging of the collection component into an object of the same shape is *order-laden*, and that the Ω story succeeds precisely by living at the one setting of the dial (powerset) where the order requirement is vacuous. The reconciliation in §7–§8 takes the order requirement seriously and localizes it, rather than wishing it away.

5 The same two equations: distributive laws and their no-go theorems

Before the reconciliation, i want to record why the collection dial cannot be turned freely if one insists on the standard monadic route to the structural connectives, because it is the same two equations again.

The structural connectives of §2, viewed monadically, ask the collection monad S to commute past the term monad T. Lifting a constructor $f : T^n \to T$ to collections requires a coherent map $(S\,A)^n \to S(A^n) \to S\,A$, and for this to be coherent across *all* the constructors of T simultaneously one wants a distributive law

$$d : S \circ T \;\Rightarrow\; T \circ S,$$

in the sense of Beck [5]. A distributive law is exactly what lets S-collections be pushed through T-structure functorially, yielding the lifted constructors.

The trouble is that such laws are scarce, and the scarcity is governed by commutativity and idempotence. The covariant powerset monad does not distribute over itself [6]; the distribution monad does not distribute over powerset [7]; and the general no-go theorems of Zwart and Marsden [8] show that once the two theories being combined carry equational shapes of the wrong kind — an idempotence or an absorption clashing with the other theory — *no* distributive law can exist. The upshot is that, along the distributive-law route, the only collection monads that commute nicely past term-like T are the set/multiset/distribution-shaped ones: exactly the ones near the top-and-commutative corner of the dial.

Principle 5.1 (One datum, read twice)**.** *The equations that obstruct the distributive law $d : ST \Rightarrow TS$ — commutativity and idempotence — are the same equations that, in §8, one must quotient* List *by to present S, and hence the same equations whose section must be postulated to package. The no-go obstruction and the packaging cost are one datum read twice: once as a law that fails, once as an operation that must be added.*

The reconciliation of §7–§8 is, in effect, a third way around the no-go theorems. The two known escapes are to *quotient* (convex powerset, indexed valuations — adjust S until the equations agree) or to *restrict* (finite, ordered — shrink until the clash cannot arise). The route here is to *add a global structuring operation* — the section — and pay for it in a named, localized axiom. §11 argues these are dual dodges.

6 The finite-limits metalanguage does exactly what we need

Now the metalanguage commitment earns its keep. Take the ambient metalanguage to be a finite-limits theory: the cartesian fragment, with $\top$, $\wedge$, $=$, and existentials only where the witness is provably unique [12, 13, 14]. Absent are $\vee$, $\bot$, $\Rightarrow$, $\neg$, unrestricted $\exists$ and $\forall$, and — in general — a subobject classifier. Three consequences, each of which turns a would-be obstruction into a feature.

(1) There is no Ω to lean on, so the syntactic route is the only route. In a topos one might hope to repair §4 with a cleverer use of the classifier. In a finite-limits metalanguage that hope is simply unavailable: a lex category need not be a topos, and the power object is not on the table. The McBride-derivative construction of §7 is therefore not *an* alternative to the Ω-story; it is the *only* story. The commitment vindicates going syntactic instead of competing with it.

(2) Choice cannot be a proposition, only an operation. In cartesian logic one cannot write the choice principle in $\forall\exists$ form: there is no $\forall$, and $\exists$ may be asserted only under uniqueness. So "adding choice" cannot mean asserting a sentence. It can only mean enlarging the signature with a function symbol and equations that make it a section of some quotient. This is precisely the shape that packaging will take in §8: pack is a new operation, and the equation $q \circ \mathsf{pack} = \mathrm{id}$ is choice-as-structure. The metalanguage forces the conjecture into its most honest and most inspectable form.

(3) No classicality collapse. In a topos, internal choice forces Boolean logic (Diaconescu [15, 16]). That argument needs $\vee$, comprehension via the power object, and effective quotients to run — *none* of which are present in the cartesian fragment. Consequently one may add packaging/choice as pure structure without collapsing the object logic to classical. Choosing finite limits is exactly what keeps choice *structural* rather than *logical*. This is the technical heart of the inspectability argument of §1: the choice we add is a nameable operation and it does not silently classicalize everything downstream.

7 The McBride derivative and the proto power object

i now build the object that replaces the power object. The move is to treat every collection *syntactically* — as a term type — and to use the McBride derivative to enumerate its splittings.

7.1 Terms, derivatives, and the zipper

Let F be a polynomial (container) endofunctor and let $T = \mu X.\, FX$ be its initial algebra, the type of terms. Write ∂F for the derivative of F: the functor of one-hole F-contexts [9, 10, 11]. The derivative of the fixpoint is a list of layers up the spine:

$$\partial T \;\cong\; \big(\partial F[T]\big)^{*} \;=\; \mathsf{List}\big(\partial F[T]\big),$$

each layer an F-node with one hole and its other children filled by subtrees. Plugging fills the hole:

$$\mathsf{plug} : \partial T \times T \longrightarrow T, \qquad \mathsf{plug}(\mathsf{nil}, t) = t, \quad \mathsf{plug}(\ell :: c, t) = \mathsf{plug}\big(c, \mathsf{fill}(\ell, t)\big).$$

Definition 7.1 (Splittings)**.** The object of *splittings* of T is

$$\mathsf{Split}(T) \;=\; \{\, (c, t) \in \partial T \times T \,\} \;\cong\; \textstyle\sum_{c:\partial T} T,$$

the pointed terms, with the whole recovered by $\mathsf{plug}(c, t)$. A splitting is a context together with a focus.

Two facts about splittings are load-bearing.

Proposition 7.2 (Positions are ordered by syntax)**.** *The positions of a term — equivalently, the contexts c appearing in its splittings — carry a canonical linear order, the traversal (Dewey) order, definable from the term's own structure by a fold. Hence ∂T is linearly ordered* per term*, with no appeal to any order on the atoms.*

Sketch. A depth-first left-to-right traversal assigns to each position a Dewey path, and Dewey paths are linearly ordered lexicographically. The assignment is a catamorphism over T, so it is definable once T is equipped with its recursor (§8); no comparison on atoms is used, because positions are indexed by constructor-argument slots, which are ordered by the arity, not by the atoms sitting in them. □

Proposition 7.3 (The aperture is the modal label)**.** *The contexts that index splittings are the same objects that label rewrites in the behavioral component of §2: both are elements of ∂T. Thus the collection component (via its splittings) and the behavioral component (via its modal labels) are glued at the derivative. In the agency vocabulary, the aperture — the locus at which a computation may be acted upon — is simultaneously the modal label and the raw material of the proto power object.*

Proposition 7.3 is the structural dividend of going through ∂T: components (1) and (3) of §2 are not merely analogous, they share their carrier.

7.2 Collections are term types too

The syntactic stance says: present the collection type itself as a term type, so the same derivative machinery applies to it. The paradigm is the list monad,

$$\mathsf{List}\,A = \mu X.\, 1 + A \times X,$$

a genuine μ-type. Its derivative in the element parameter is the pair of a prefix and a suffix,

$$\partial_A(\mathsf{List}\,A) \;\cong\; \mathsf{List}\,A \times \mathsf{List}\,A,$$

so a splitting of a list is a triple $(\text{prefix}, \text{focus}, \text{suffix})$ — exactly a choice of one element to keep in view, with its surroundings. Gathering foci across a family of splittings and re-collecting rebuilds a *sub*collection. This is the concrete content of "no list of all lists": each individual sublist is free to form, but assembling all of them into one list requires ordering them.

Definition 7.4 (Proto power object)**.** For a term type T, the *proto power object* $\Pi(T)$ is the container of splittings, written $(\partial T, T)$: shapes are contexts $c : \partial T$, and the position at each shape is the focus of type T. Its extension is $\Pi(T)(X) = \sum_{c:\partial T} X$, and $\Pi(T)(T) = \mathsf{Split}(T)$.

$\Pi(T)$ is "proto" because it enumerates one-step focusings; it is not yet closed under gathering foci into a subcollection of the same shape as T. Closing it is packaging.

8 Packaging as a section, and the choice conjecture

8.1 Presenting collections as quotients of the free case

Present each collection monad S syntactically, as a quotient of the free (list) collection:

$$\mathsf{List}\,A \xrightarrow{q_S} S\,A,$$

where

$$q_{\mathsf{List}} = \mathrm{id}, \qquad q_{\mathsf{Bag}} = (-)/\mathrm{perm}, \qquad q_{\mathsf{P}_{\mathrm{fin}}} = (-)/(\mathrm{perm} + \mathrm{idem}).$$

List is the free monoid — the syntactic collection, paying nothing. Bag adds commutativity (permutation). $\mathsf{P}_{\mathrm{fin}}$ adds commutativity and idempotence. These are precisely the two coordinates of the dial (Principle 3.1) and the two culprits of the no-go theorems (Principle 5.1).

8.2 Two additions before choice

Two things must be added to the bare finite-limits base before packaging can even be discussed, and honesty requires naming them.

Remark 8.1 (The recursor is the first non-Horn addition)**.** Having "the type of terms" means an initial algebra μF, and using it — defining any fold, hence pack, hence the Dewey order of Proposition 7.2 — requires its *recursor*. Formation and introduction for a μ-type are algebraic; elimination (induction/recursion) is not cartesian. So the recursor is the first thing added to the bare base, and we add the *weakest* one that still defines pack: a non-dependent catamorphism. Everything downstream, including packaging, is a fold and inherits exactly this recursor and no more.

Remark 8.2 (Quotients before sections)**.** For non-free S, the map q_S is a coequalizer, which a bare lex category may lack. Presenting S therefore already commits one to effective quotients (pretopos-flavored structure) *before* asking them to split. Packaging then splits what one was forced to construct.

So the layering is: cartesian base $\to$ weakest recursor for T (Remark 8.1) $\to$ effective quotients for non-free S $\to$ the section below. Choice sits only at the top layer.

8.3 The packaging axiom

Definition 8.3 (Packaging)**.** A *packaging* for S is a signature operation

$$\mathsf{pack}_S : S\,A \longrightarrow \mathsf{List}\,A \qquad \text{with} \qquad q_S \circ \mathsf{pack}_S = \mathrm{id}_{SA},$$

a section of the presenting quotient: a choice, for each S-collection, of a list representative. The axiom asserting the existence of pack_S is PACK_S. In the finite-limits metalanguage PACK_S is not a proposition but the enlargement of the signature by pack_S and the above equation — choice-as-structure, per §6(2).

Definition 8.3 is the syntactic form of the closure operation on the proto power object: to assemble the splittings of $\Pi(T)$ (Definition 7.4) into a S-collection of the same shape is to apply q_S, and to make that assembly canonical — to build $T[(\partial T, T)]$, the collection whose atoms are splittings — is to choose the section pack_S.

8.4 The choice conjecture, parametric in the atom order

Theorem 8.4 (Derivability of packaging, per instance)**.** *Modulo the weakest recursor of Remark 8.1, and for the section taken uniformly (natural in A):*

(i) $\mathsf{PACK}_{\mathsf{List}}$ *is derivable outright;* $\mathsf{pack}_{\mathsf{List}} = \mathrm{id}$.

(ii) $\mathsf{PACK}_{\mathsf{Bag}}$ *is derivable if the atom object A carries a linear order $\leq$ definable in the theory — take* $\mathsf{pack}_{\mathsf{Bag}} = \mathsf{sort}_{\leq}$ *— and, conversely, a uniform section of q_{Bag} determines a linear order on A (read off from its action on two-element bags). So* $\mathsf{PACK}_{\mathsf{Bag}}$ *is inter-derivable with "A is linearly ordered."*

(iii) $\mathsf{PACK}_{\mathsf{P}_{\mathrm{fin}}}$ *is derivable if A is linearly ordered* and *equality on A is available to deduplicate — take* $\mathsf{pack}_{\mathsf{P}_{\mathrm{fin}}} = \mathsf{dedup} \circ \mathsf{sort}_{\leq}$*; conversely a uniform section determines both an order and a canonical representative.*

Sketch. (i) is immediate. For (ii), $\mathsf{sort}_{\leq}$ is an insertion sort, a fold over $\mathsf{List}\,A$ using $\leq$, hence definable from the weakest recursor and $\leq$; it lands in a canonical (sorted) representative of each permutation class, so $q_{\mathsf{Bag}} \circ \mathsf{sort}_{\leq} = \mathrm{id}$. Conversely, given a uniform section s, its value on the bag $\{a, b\}$ is one of the lists $[a, b]$ or $[b, a]$; uniformity (naturality in A) forces this decision to cohere into a total order, and the monoid equations force transitivity. (iii) adds idempotence: a section of the combined quotient must additionally select, from each finite subset, a duplicate-free list, i.e. a decision of equality plus the order of (ii). $\square$

Theorem 8.4 localizes the choice content exactly. Read against the earlier, topos-flavored heuristic in which ordered collections were the expensive case (linearize an unordered carrier =

well-order = full choice), it *inverts* the cost: here List is free and the quotients are dear. The two framings concern different maps, and the finite-limits, syntactic stance is what reconciles them:

Principle 8.5 (The McBride move trades and discharges)**.** *Going through the derivative trades the* well-ordering flavor *of choice for the* quotient-section flavor*, and then discharges the former for free. The foci of a term-splitting are* positions*, and positions are pre-ordered by the term's own structure (Proposition 7.2). Syntax hands you the linear order. So over a term rewrite system, packaging is derivable from the positional order, and choice re-enters only through the quotient equations, and only when the atom object A carries no cartesian-definable linear order.*

Conjecture 8.6 (Choice content of packaging)**.** *The choice content of* PACK_S *equals the assertion that the atom object A admits a linear order. Equivalently: with an order on A supplied,* PACK_S *is derivable for every finitary S presented as a quotient of* List*; without one,* PACK_S *is the added choice operation, and its strength is graded by the equations q_S quotients* List *by. In particular the construction is kept* parametric in an order on atoms*: supplying the parameter makes packaging free; withholding it isolates the entire choice residue at the atom object.*

The parametricity is deliberate and is the hook for later work (§11): in a reflective setting the atoms are quoted processes — names — and an order on names is exactly a place where a knot may be tied. Leaving the order parametric localizes the choice content at the reflection boundary, where we will want to act on it.

9 Grading the conjecture in the Weihrauch lattice

Conjecture 8.6 is not a switch but a dial, and the natural home for its grading is the Weihrauch lattice [17], which measures the uniform strength of a choice principle. i state the intended grading as a program; the finitary cases are within reach of the sketch of Theorem 8.4, the infinitary ones are open.

Conjecture 9.1 (Weihrauch grading)**.** *The Weihrauch degree of* PACK_S *over an atom object A is monotone in the substructural strength of S — i.e. in the equations q_S adds to* List *— and is bounded as follows:*

- $\mathsf{PACK}_{\mathsf{List}}$*: the identity; the bottom of the lattice.*
- $\mathsf{PACK}_{\mathsf{Bag}}$ *over finite bags on discrete A: the finite/bounded choice needed to order each finite support uniformly; low in the lattice, below* LPO *when A is discrete.*
- $\mathsf{PACK}_{\mathsf{Bag}}$ *over an arbitrary (possibly infinite) atom object: a closed-choice principle* C_A *on the carrier, climbing toward well-ordering as A grows; the well-ordering flavor of Principle 8.5 re-emerges exactly here, when the syntactic order is unavailable.*
- $\mathsf{PACK}_{\mathsf{P}_{\mathrm{fin}}}$*: the degree of* $\mathsf{PACK}_{\mathsf{Bag}}$ *joined with the equality/deduplication decision, adding an* LPO*-flavored component for the idempotence quotient.*

The value of the Weihrauch reading is that it reframes the no-go theorems of §5 not as yes/no obstructions but as *choice-strength measurements*. Where the classical no-go statement says "no distributive law exists," the graded statement says "the section that stands in for the missing law has degree d," and d is a monotone invariant of how far S sits from the free case. That is the stronger thesis hiding inside the bare conjecture.

10 The three instances, concretely

i collect the worked instances. In each, the theory is the cartesian base extended by the weakest recursor for the relevant μ-type (Remark 8.1); the atom order is the parameter of Conjecture 8.6.

S	equations added to List	pack_S	choice residue
List	none	id	none
Bag	commutativity	$\mathsf{sort}_{\leq}$	order on A
$\mathsf{P}_{\mathrm{fin}}$	commutativity, idempotence	$\mathsf{dedup} \circ \mathsf{sort}_{\leq}$	order on A, $=$ on A

Example 10.1 (List: nothing is spent)**.** $\mathsf{List}\, A = \mu X.\, 1 + A \times X$; $q_{\mathsf{List}} = \mathrm{id}$; the section is the identity. The proto power object $\Pi(\mathsf{List}\, A)$ has $\partial(\mathsf{List}\, A) \cong \mathsf{List}\, A \times \mathsf{List}\, A$ (prefix, suffix), and a splitting is $(\mathrm{prefix}, \mathrm{focus}, \mathrm{suffix})$. Packaging is free because there is nothing to reassemble past the given list order.

Example 10.2 (Bag: order on atoms is spent)**.** $\mathsf{Bag}\, A = \mathsf{List}\, A/\mathrm{perm}$. With $\leq$ on A, insertion sort is a fold selecting the sorted representative, and $q_{\mathsf{Bag}} \circ \mathsf{sort}_{\leq} = \mathrm{id}$. Withhold $\leq$ and no cartesian term produces a representative uniformly; $\mathsf{pack}_{\mathsf{Bag}}$ must be postulated, and by Theorem 8.4(ii) its postulation is inter-derivable with linearly ordering A. This is the cleanest witness for Conjecture 8.6.

Example 10.3 (Finite powerset: order and equality)**.** $\mathsf{P}_{\mathrm{fin}}\, A = \mathsf{List}\, A/(\mathrm{perm}{+}\mathrm{idem})$. Packaging needs the sorted representative of the previous example *and* a deduplication, hence decidable equality on A in addition to the order. The two additions are exactly the two quotient equations, confirming Principle 5.1: the idempotence that obstructs $\mathsf{P}_{\mathrm{fin}}$'s distributive laws is the same idempotence that must be paid for, as dedup, at packaging time.

11 Discussion

No classicality collapse. The absence of Diaconescu's argument in the cartesian fragment (§6(3)) is what makes this entire program viable. We add a choice operation — pack_S — and the object logic does not thereby go Boolean, because the collapse needs $\vee$, comprehension, and effective-quotient machinery that the metalanguage does not provide. Choice remains structural. A logic assembled this way can have any setting of the substructural dial (Principle 3.1) and still admit packaging, without being forced to classicalize. This is the payoff i most want the AI community to register: inspectable choice that does not contaminate the logic it serves.

Quotient versus choice as dual dodges. §5 noted two standard escapes from the no-go theorems — quotient (adjust S until the equations agree) and restrict (shrink until they cannot clash) — and offered a third, adding a global section. i conjecture these are dual: the quotient-dodge builds a coequalizer, the section-dodge splits it; the restrict-dodge builds a well-founded (finite, ordered) carrier, the section-dodge well-orders it. In slogan form: *coequalizer versus Zorn.* Making this duality precise — exhibiting the section-dodge as the formal dual of the quotient-dodge in the appropriate fibration — is the natural next technical step, and it would situate the construction cleanly against the existing distributive-law literature.

Graphs are the genuine frontier. Everything here rests on T being a polynomial/container functor, so that ∂T exists and Definition 7.4 is available. Lists and trees are fine. Graphs are not free/regular in the naive sense, the McBride derivative does not directly apply, and the proto power object is not defined. i bracket the graph case explicitly rather than let it ride along with lists and trees; adapting the derivative to non-regular carriers (or replacing it by a suitable comonadic/traced structure) is the honest open problem for extending the collection dial past the container-shaped settings.

The atom-order parameter and knot-tying. Conjecture 8.6 keeps the order on atoms parametric on purpose. In the reflective (rho) setting the atoms are quoted processes — names — and the boundary between process and name is exactly the reflection boundary. An order on names is therefore a structure imposed *at* that boundary, and it is precisely the datum a knot-tying construction will manipulate. By isolating the entire choice residue of packaging in the atom order, the present construction leaves the knot exactly one place to be tied: supply the order (and packaging trivializes; the knot is untied) or withhold it (and the choice residue is the knot). This is the sense in which the finite-limits, derivative-based account is not merely a reconciliation of the collection component with categorical logic, but a setup for the reflective constructions to come.

Summary. The three-component decomposition makes the seams of a logic visible under realizability; the collection component is a substructural dial governed by commutativity and idempotence; those same two equations obstruct the distributive laws and set the cost of packaging; the finite-limits metalanguage removes the subobject classifier (forcing the syntactic route) and forbids stating choice as a proposition (forcing it to be a named section), while blocking the classicality collapse that would otherwise make this untenable; the McBride derivative supplies the proto power object and, through the positional order on splittings, discharges the well-ordering flavor of choice; and what remains — the choice content of packaging — is exactly the order on atoms, kept parametric for the knot to act on. That is the construction i wanted to formalize, and the shape of the program it opens.

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